

Case Report: Use of Sports and Performance Vision Training to Benefit a Low Vision Patient's Function

Daniel M. Laby, MD*

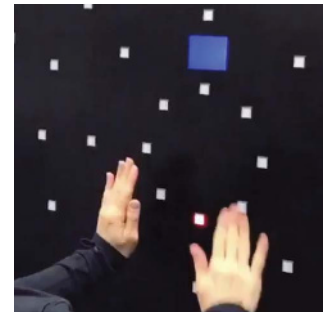
SIGNIFICANCE: Despite our inability to attenuate the course of many ocular diseases that can ultimately lead to loss or significantly decreased visual function, this report describes a potential technique to aid such patients in maximizing the use of the vision that remains.

PURPOSE: The aim of this study was to demonstrate the applicability of utilizing sports vision training to improve objective and subjective visuomotor function in a low vision patient.

CASE REPORT: A 37-year-old woman with Usher syndrome presented with reduced central visual acuity and visual field. Although we were unable to reverse the damage resulting from her diagnosis, we were able to improve the use of the remaining vision. A 27 to 31% improvement in hand-eye coordination was achieved along with a 41% improvement in object tracking and visual concentration. Most importantly, following the 14-week training period, there was also a subjective improvement in the patient's appreciation of her visual ability.

CONCLUSIONS: The sports vision literature cites many examples in which sports vision training is useful in improving visuomotor and on-field performance. We hypothesized that these techniques may be used to aid not only athletes but also patients with low vision. Despite suffering from reduced acuity and a limited visual field, these patients often still have a significant amount of vision ability that can be used to guide motor actions. Using techniques to increase the efficient use of this remaining vision may reduce the impact of the reduced visual function and aid in activities of daily living.

Optom Vis Sci 2018;95:898–901. doi:10.1097/OPX.0000000000001231
Copyright © 2018 American Academy of Optometry



Author Affiliations:

State University of New York,
College of Optometry, New York,
New York
*drlaby@sportsvision.nyc

Usher syndrome is one of the most common conditions that affect both vision (retinitis pigmentosa) and hearing. Much work has been done regarding the visual search and gaze behaviors of retinitis pigmentosa patients, with techniques used to improve real-world function despite vision loss. Turano et al.^{1,2} noted patients with advanced visual impairment sample their environment differently and require more mental effort for walking compared with fully sighted individuals. Geruschat et al.³ found retinitis pigmentosa patients travel more slowly than normally sighted individuals, and their self-reported, mobility-related behaviors were significantly correlated to walking speed. Timmis et al.⁴ found retinitis pigmentosa individuals had more active visual search patterns, whereas Vargas-Martin and Peli⁵ found subjects with severe peripheral field loss did not increase their scanning eye movements to compensate for missing peripheral information. In light of these findings, several interventions may aid patients with retinitis pigmentosa to more successfully navigate their environment. For example, Peli et al.⁶ used mathematical modeling of visual field expansion devices (e.g., prisms) to reduce the risk of unintended collisions. Ivanov et al.⁷ used saccadic training and noted that, following training, response times improved, as did preferred walking speed.

We report a case of Usher syndrome in which vision and hearing loss has progressed since age 12 years. Although there is no way to reverse sensory vision loss, we applied sports vision training to enhance the usefulness of the remaining vision.

CASE REPORT

A 37-year-old woman requested assistance with vision. Being an amateur athlete, she was motivated to make the best use of her remaining vision. Medical history revealed that she has had vision difficulties since age 12 years. She also has had hearing difficulty since childhood and has recently required cochlear implants. Genetic testing revealed a homozygous *N48k* mutation in the *USH3A* locus for Usher syndrome type 3.

Functionally, she drove until her late 20s. Her best corrected vision is approximately 20/50 in each eye. Anterior segment evaluation is normal, except for mild posterior subcapsular cataracts in each eye and mild cortical opacities in the left eye. Fundus evaluation reveals pale optic discs bilaterally, with vascular attenuation and atrophic/pigmentary changes in the macular region of each eye, as well as typical bone spicule-like pigmentation. Goldmann visual field reveals a constricted field diameter (using a V4e target) of 20 degrees in the right eye and 26 degrees in the left eye. Several isolated V4e islands were noted temporally in each eye. She was previously evaluated at the vision rehabilitation clinic at the University of Iowa, and she has tried several techniques to improve her visual function including magnification at near, reversed contrast electronic devices, and Corning Photochromic Filters.

Following sports vision assessment, it was determined that the greatest deficits were in hand-eye coordination/reaction to stimuli

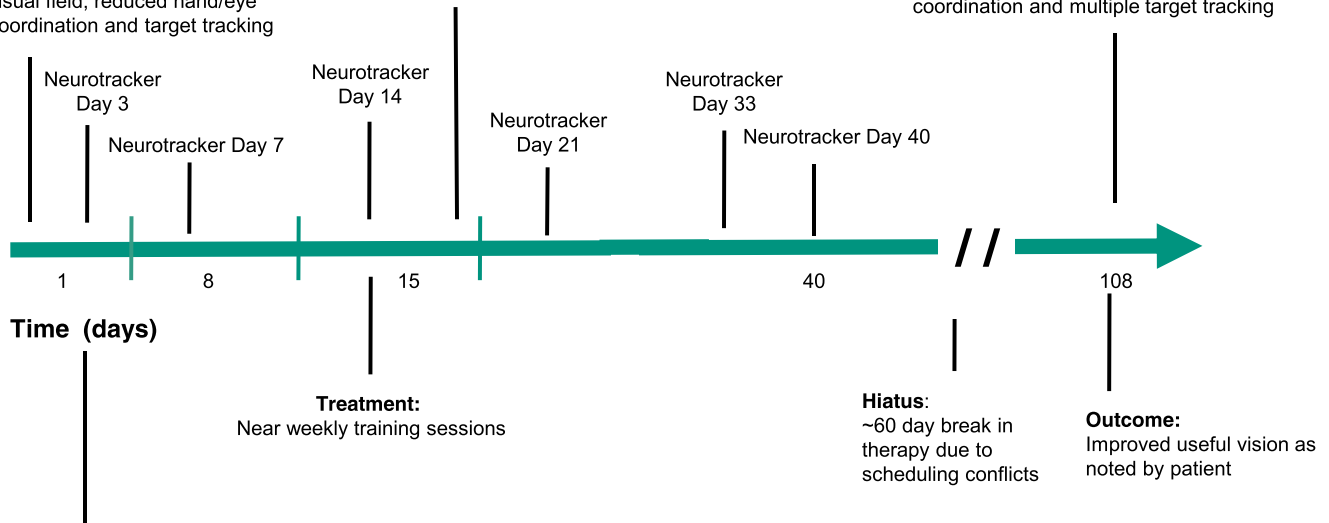
Symptoms / Diagnosis

Initial Assessment:

Decreased visual acuity and visual field, reduced hand/eye coordination and target tracking

Weekly Training Sessions

Final Sports Vision Training Session and assessment: Improved hand/eye coordination and multiple target tracking



Begin Sports Vision Training:

Dynavision and Neurotracker Systems

Interventions / Outcome

FIGURE 1. Clinical training timeline: a 37-year-old white woman diagnosed with Usher syndrome type 3. Neurotracker training was performed on eight of the visits. Dynavision training was performed on each of the 12 visits.

and concentration on changing visual stimuli. A weekly training program was devised to aid these visual skills. The Dynavision (D2) system was chosen for hand-eye coordination/reaction training, whereas the Neurotracker system was used for visual concentration.

Two different modes, following the manufacturer's recommended procedure, were used, each utilizing the central ring of individual lights on the D2 board. Initially, a "proactive" mode was used, followed each session by "reactive" training. In the proactive

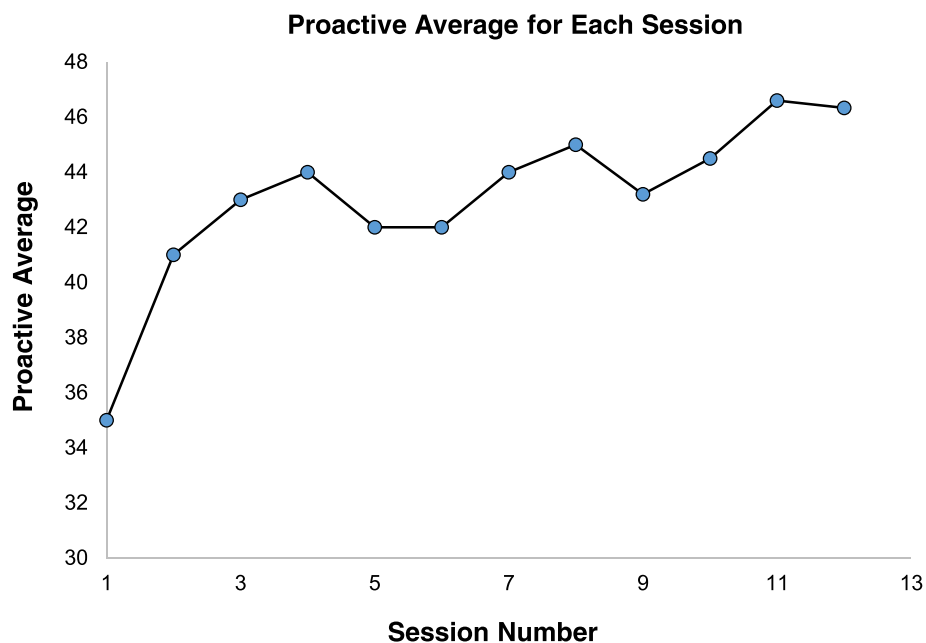


FIGURE 2. Average Dynavision proactive scores for each of the 12 training sessions. Note the rapid increase over the initial four sessions, followed by a slower progression during the remaining sessions.

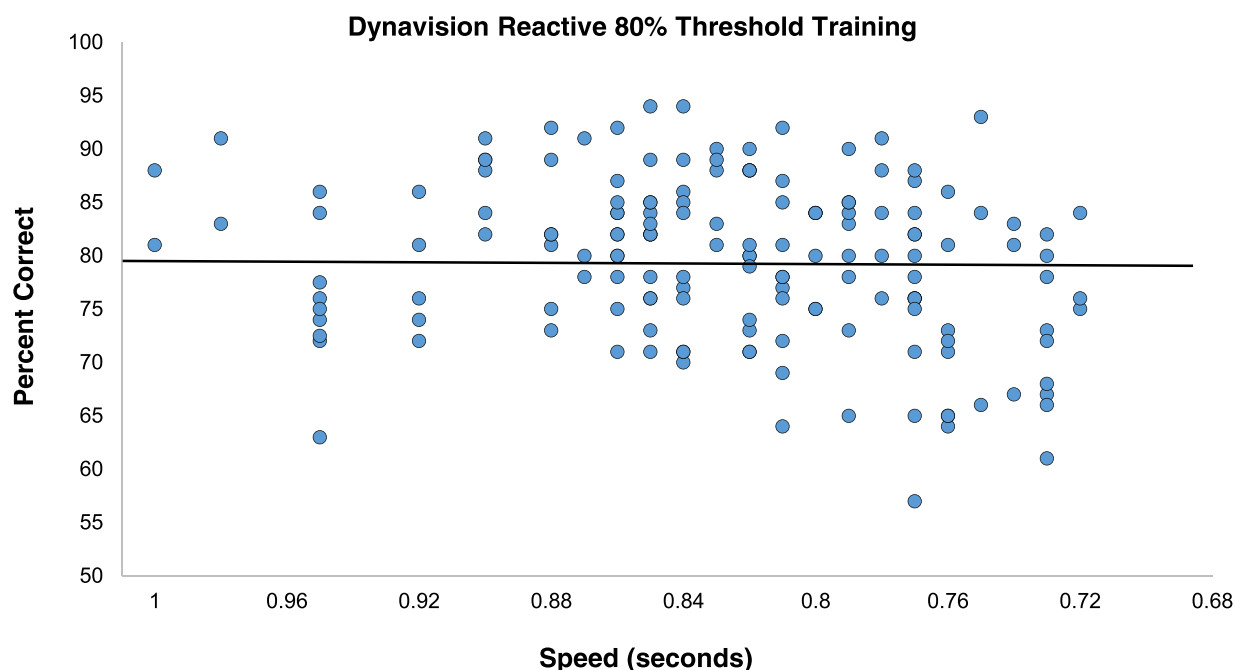


FIGURE 3. Dynavision reactive training results beginning with an initial presentation duration of 1 second and ending at 0.72 seconds at the final training session. Note the 80% correct threshold (dark line). In order to reduce the presentation duration, two trials of better than 80% success must be achieved. At the beginning of each training session, the last several presentation durations were repeated, accounting for the multiple results greater than the 80% threshold for most duration speeds.

mode, a single button light is lit and remains lit until the subject presses the lit button. Immediately following, another random button is lit, again staying lit until pressed. This is repeated for 30 seconds. The total number of buttons pressed in the 30-second testing period is recorded as the proactive score. In the reactive mode, buttons are lit and then turned off at a pre-set interval (beginning at 1000 milliseconds). The subject is asked to press the button before the light is turned off. Once one button turns off, another is randomly lit for the same pre-set duration. The percentage of successfully pressed buttons is recorded as the reactive score. Once the subject scores 80% or better on two occasions, the lit interval is shortened by 10 to 30 milliseconds, in order to maintain task difficulty and gain further improvement. The subject was oriented to the board at arm's length and at eye level, but was not restricted in movement, resulting in variable visual field target location.

The Neurotracker system requires tracking a single ball in a field of eight balls as they move randomly within a virtual three-dimensional cube viewed through a head-mounted display consisting of 46 degrees of visual angle at the level of the target screen. The balls move for 8 seconds, and upon stopping, the previously marked ball must be reidentified by the subject. Each time the ball is correctly identified, the movement speed of the balls is increased, whereas following an incorrect identification the speed is decreased. The threshold score (core score) is automatically calculated by the Neurotracker CORE software, following 20 trials, based on the fastest ball speed with consistently correct identification.

Our subject presented for 12 sessions over a 14-week period (Fig. 1). Average Dynavision proactive training began with a score of 35 buttons pressed and ended with 46 buttons pressed, a 31% increase (Fig. 2), whereas reactive ability improved from a

beginning duration of 1000 milliseconds, ending at 720 milliseconds, a 28% improvement (Fig. 3). Neurotracker ability began with a "core score" of 2.30 and ended with 3.25, a 41% improvement (Fig. 4). More importantly, subjective improvement in visual function in terms of both "quickness" of seeing objects in daily living activities and an increased ability to avoid obstacles was noted by the subject. Specifically, feeling more "energetic" and "focused" and decreased collisions with objects were noted following training.

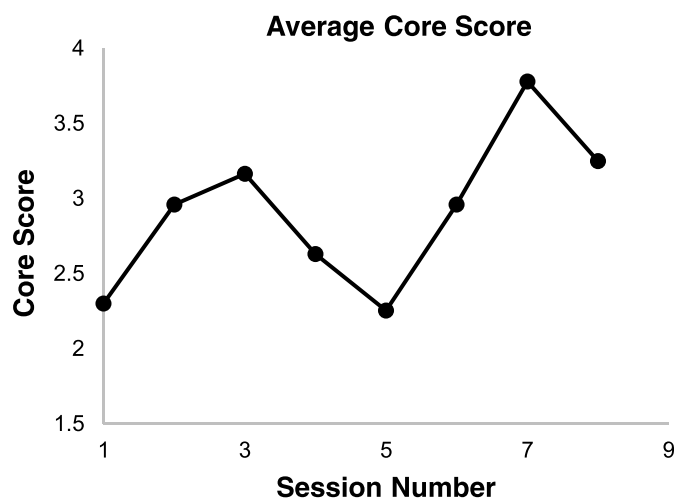


FIGURE 4. Average Neurotracker core score results over eight training sessions. Because of time constraints, core score testing was not performed at each training session.

DISCUSSION

Sports and performance vision is a relatively new subspecialty of eye care. Recently, there has been increased interest in how this approach can improve athletic performance,⁸ as well as other tasks.

In terms of athletic performance, the importance of the visual system is intuitive. Regardless of the sport, visual information must be used to coordinate a motor action in order to succeed. In addition to the classic visual system, sports vision specialists evaluate how that visual signal is used to produce a desired motor action.

To understand the role of higher cortical processes in athletic performance, one should consider the “bottleneck” concept. Our retinas sense a great deal of visual information, which is transmitted to the visual cortex, or higher cortical regions. All visual information cannot be appreciated simultaneously, and there is a need to prioritize which information is used in order to allow an optimal motor response. The system would simply be overwhelmed if a filter was not applied to reduce the amount of information and focus attention on that which is critical to the performance of a task.⁹ Gao et al.¹⁰ noted that athletes had superior visuomotor skills compared with nonathletes. Their report suggests that athletes who can more efficiently determine which visual information is useful are more likely to be successful.

It is precisely the above that all visual information is not required to perform well, but instead it is more important to have the ability to

identify which visual data are needed to create an efficient and optimal visual-motor response, which led us to consider the possibility of sports vision training as a potentially useful intervention in this patient with Usher syndrome.

Although the amount and quality of visual information may be relatively reduced in Usher syndrome, as compared with subjects with a normal retina, the ability to more efficiently process the remaining information and create an improved motor response might allow those with reduced vision to perform at an improved level.

Based on previous reports,¹⁰ we elected to train hand-eye coordination and multiple-object tracking because these were likely the most critical abilities for our subject. Although required to begin at a relatively easier level than most athletes, as a result of the limited visual ability present, our subject quickly advanced and improved in ability in each of these tasks.

In patients with reduced visual ability, sports vision training may provide an additional intervention allowing for improved use of remaining visual ability. In this case, improved ability appears to have led to improved daily living as self-reported by our subject. This case report suggests that sports vision training may be an appropriate adjunct therapy for low vision patients and adds another option for caregivers in working with low vision patients. Additional study with larger cohorts, different visual deficits, and additional training interventions will be needed in order to fully understand the role of this emerging specialty in the care of low vision patients.

ARTICLE INFORMATION

Submitted: October 14, 2017

Accepted: February 18, 2018

Funding/Support: The author has reported no funding/support.

Conflict of Interest Disclosure: The author has reported no financial conflict of interest.

Author Contributions: Conceptualization: DML; Data Curation: DML; Formal Analysis: DML; Investigation: DML; Methodology: DML; Project Administration: DML; Validation: DML; Visualization: DML; Writing – Original Draft: DML; Writing – Review & Editing: DML.

REFERENCES

1. Turano KA, Geraschat DR, Baker FH, et al. Direction of Gaze While Walking a Simple Route: Persons

with Normal Vision and Persons with Retinitis Pigmentosa. *Optom Vis Sci* 2001;78:697–75.

2. Turano KA, Geraschat DR, Stahl JW. Mental Effort Required for Walking: Effects of Retinitis Pigmentosa. *Optom Vis Sci* 1998;75:879–86.

3. Geraschat DR, Turano KA, Stahl JW. Traditional Measures of Mobility Performance and Retinitis Pigmentosa. *Optom Vis Sci* 1998;75:525–37.

4. Timmis MA, Allsop J, Baranian M, et al. Visual Search Behavior in Individuals with Retinitis Pigmentosa during Level Walking and Obstacle Crossing. *Invest Ophthalmol Vis Sci* 2017;58:4737–46.

5. Vargas-Martin F, Peli E. Eye Movements of Patients with Tunnel Vision While Walking. *Invest Ophthalmol Vis Sci* 2006;47:5295–302.

6. Peli E, Apfelbaum H, Berson EL, et al. The Risk of Pedestrian Collisions with Peripheral Visual Field Loss. *J Vis* 2016;16:5.

7. Ivanov IV, Mackeben M, Vollmer A, et al. Eye Movement Training and Suggested Gaze Strategies in Tunnel Vision—A Randomized and Controlled Pilot Study. *PLoS One* 2016;11:e0157825.

8. Appelbaum LG, Erickson G. Sports Vision Training: A Review of the State-of-the-art in Digital Training Techniques. *Int Rev Sport Exerc Psychol* 2016; doi 10.1080/1750984X.2016.1266376.

9. Tombu MN, Asplund CL, Dux PE, et al. A Unified Attentional Bottleneck in the Human Brain. *Proc Natl Acad Sci U S A* 2011;108:13426–31.

10. Gao Y, Chen L, Yang SN, et al. Contributions of Visuo-oculomotor Abilities to Interceptive Skills in Sports. *Optom Vis Sci* 2015;92:679–89.